

Far-field boundary conditions for an immersed-boundary compressible solver: reflection, mean-state anchoring, and a drift-free aeroacoustic metric

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¹Cerisse compressible immersed-boundary solver, validation campaign

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The Cerisse Cartesian-AMR immersed-boundary solver offers three usable far-field ghost-cell closures — zero-gradient extrapolation (*foextrap*), hard Dirichlet freestream, and an algebraic Riemann-invariant “NSCBC” — and the question of which best represents an open boundary recurs in every external-flow validation case. We answer it with three controlled experiments of increasing dimensionality. A one-dimensional Poinso–Lele acoustic-pulse benchmark shows the as-shipped algebraic closure returns 78% of a normally incident wave; we trace this exactly to the $\sigma \rightarrow \infty$ (perfectly reflecting) limit of a relaxed characteristic boundary condition, and a one-line correction ($J_{\text{ghost}}^- = J_{\infty}^-$ instead of $p_{\text{ghost}} = p_{\infty}$) restores $R < 10^{-5}$ while preserving mean-state anchoring. A two-dimensional co-rotating vortex pair then tests genuine multi-directional radiation. We introduce a drift-free, phase-aware acoustic-corruption metric — a per-point least-squares single-tone phasor evaluated in a far-field annulus and referenced to a reflection-free wide-domain truth, with a truth-against-itself noise floor — and find that *every* closure’s corruption of the radiated quadrupole lies below the measurement floor: the free-radiation wave does not discriminate the boundary conditions. The single field that does separate them is the mean pressure (the DC mode). Finally, a supersonic cylinder demonstrates where the characteristic closure is decisive: when an oblique bow shock crosses a near lateral boundary, the corrected characteristic condition enables a $2.5\times$ domain reduction with the shock stand-off preserved to 7%, whereas *foextrap* catastrophically destroys the upstream field. Boundary-condition quality for compressible external flow is therefore governed by mean-state anchoring and shock crossing, not by free-wave reflection.

Key words: computational methods, shock waves, aeroacoustics

1. Introduction

The Cerisse solver is a block-structured Cartesian adaptive-mesh-refinement (AMR) code for the compressible Navier–Stokes equations with an image-point immersed-boundary method (IBM); it uses a WENO-Z fifth-order reconstruction, an HLLC Riemann solver and a strong-stability-preserving Runge–Kutta integrator. Its external-flow validation suite — a cylinder at $M = 1.7$, spheres at $M = 2$ and $M = 4$ — prescribes a supersonic Dirichlet inflow on the upstream face and one of several closures

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on the remaining (lateral and downstream) faces. A schlieren A/B comparison on the cylinder showed visible differences in the lateral far field, raising a question that this paper settles: *which far-field boundary condition (BC) best represents an open boundary for compressible external flow, and why?*

The literature on non-reflecting boundaries is mature. Thompson (1987) cast the Euler equations in characteristic form at boundaries; Poinso & Lele (1992) extended this to the Navier–Stokes characteristic boundary condition (NSCBC) with a linear-relaxation closure for the incoming wave, building on the acoustically non-reflecting analysis of Rudy & Strikwerda (1980). Multidimensional corrections that account for transverse and viscous terms were added by Yoo & Im (2007) and Lodato *et al.* (2008). Colonius (2004) reviews the field, including absorbing (sponge) layers. Crucially, Motheau *et al.* (2017) showed that a *ghost-cell* NSCBC — the only kind expressible in a finite-volume AMR code that fills ghost cells once per stage — is a legitimate and modern formulation, not a compromise.

Against this backdrop we report three findings. First (§§2–3), a 1-D benchmark diagnoses why the solver’s algebraic “NSCBC” reflected, and a corrected characteristic closure is verified non-reflecting. Second (§§4–6), a 2-D co-rotating vortex pair, analysed with a new drift-free aeroacoustic metric, shows that for free radiation no closure beats simple extrapolation on the wave. Third (§7), a supersonic cylinder identifies the regime — oblique strong-shock crossing — where the characteristic closure is genuinely and dramatically superior. Throughout, quantities are non-dimensionalised by a reference density, sound speed and length.

2. The far-field closures

Let $\mathbf{n}$ be the outward unit normal and $\mathbf{U} = (\rho, \rho\mathbf{u}, \rho E)^\top$ the conservative state. Cerisse fills ghost cells through a user `bcnormal` routine; the integer codes 1 (`ext_dir`), 2 (`foextrap`) and 7 (`ext_dir`, `characteristic`) select the closures below. With $u_n = \mathbf{u} \cdot \mathbf{n}$ and sound speed c the boundary eigenvalues are $\{u_n - c, u_n, u_n + c\}$ and four regimes arise.

2.1. Zero-gradient extrapolation (*foextrap*, code 2)

First-order extrapolation copies the first interior cell into every ghost layer,

$$\mathbf{U}_{\text{ghost}} = \mathbf{U}_{\text{int}} \iff \partial\mathbf{U}/\partial n|_{\partial\Omega} = \mathbf{0}. \quad (2.1)$$

Nothing is prescribed; every characteristic is extrapolated. For a purely outgoing disturbance the ghost is a faithful continuation, so the boundary Riemann problem has (to first order) no jump and no reflected wave. The weakness is that nothing anchors the boundary, so in a closed subsonic region the mean pressure can drift, and obliquely incident waves are partially reflected.

2.2. Hard Dirichlet freestream (code 1)

Every ghost layer is overwritten with the constant freestream $\mathbf{U}_{\text{ghost}} = \mathbf{U}_\infty$. At a supersonic inflow ($M_n \leq -1$) all characteristics enter and this is exact; at a subsonic boundary it over-specifies, clamping $\rho, \mathbf{u}$ to freestream and corrupting any outgoing entropy or vorticity wave, and conflicting with any non-freestream mean state (e.g. post-shock gas where a bow shock crosses the boundary).

2.3. Algebraic Riemann-invariant closure (“NSCBC”, code 7)

The ghost is built from a local characteristic analysis. For the subsonic outflow branch $0 \leq u_n < c$ the outgoing invariant $J^+ = u_n + 2c/(\gamma - 1)$ and the entropy $s = p/\rho^\gamma$ are

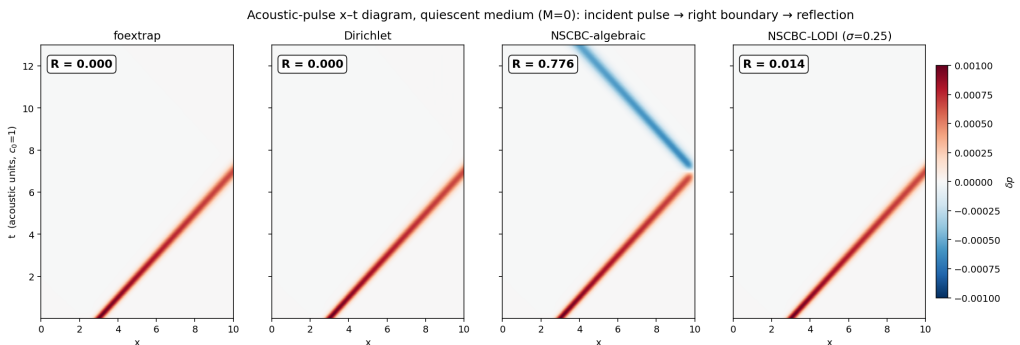


FIGURE 1. Acoustic-pulse x - t diagrams at $M = 0$. The incident pulse travels up-and-right; a returning (up-and-left) streak is a reflection. Only the as-shipped algebraic “NSCBC” (2.2) produces one; *foextrap*, hard Dirichlet and the relaxed reference let the pulse leave cleanly.

taken from the interior, and the single incoming acoustic characteristic must be fixed. The as-shipped closure *hard-imposed the back-pressure*,

$$\boxed{p_{\text{ghost}} = p_{\infty}}, \quad \rho, \mathbf{u}, s \text{ from interior.} \quad (2.2)$$

This is the standard steady-state external-aerodynamics far-field: at convergence of a steady flow no waves reach the boundary and it is exact. For *unsteady* waves, however, (2.2) is reflective, as §3 shows.

2.4. Reference: relaxed Poinot–Lele NSCBC

The proper NSCBC does not pin the pressure. The outgoing amplitudes are computed from interior one-sided differences and the single incoming amplitude $\mathcal{L}_1$ is *relaxed* toward the target,

$$\mathcal{L}_1 = \sigma (1 - M^2) \frac{c}{L} (p - p_{\infty}), \quad (2.3)$$

with $\sigma \approx 0.25$ and L a reference length, after which the boundary state evolves by the LODI relations

$$\frac{\partial p}{\partial t} = -\frac{1}{2}(\mathcal{L}_5 + \mathcal{L}_1), \quad \frac{\partial u_n}{\partial t} = -\frac{1}{2\rho c}(\mathcal{L}_5 - \mathcal{L}_1). \quad (2.4)$$

The limit $\sigma \rightarrow 0$ is perfectly non-reflecting but lets p drift; $\sigma \rightarrow \infty$ forces $p = p_{\infty}$ instantly and is perfectly reflecting — i.e. (2.2).

3. One-dimensional reflection benchmark

We solve the 1-D Euler equations with the same HLLC solver Cerisse uses (first-order Godunov, $N = 1000$, $L = 10$, $c_0 = 1$). A right-running acoustic pulse $\delta p = A \exp[-((x - x_0)/w)^2]$ with $\delta u = \delta p/(\rho_0 c_0)$, $\delta \rho = \delta p/c_0^2$ is launched toward the outflow boundary in a uniform mean flow of normal Mach number M_n . The ghost BCs are filled exactly as in the solver; the reference (2.3) replaces the boundary-cell right-hand side by the LODI update. The energy-based reflection coefficient $R = \max_t |\delta p_{\text{ref}}| / \max_t |\delta p_{\text{inc}}|$ is measured at an interior sensor.

3.1. Reflection at a quiescent boundary

Figure 1 shows the x - t evolution. Table 1 lists R at $M = 0$: the algebraic closure returns $R = 0.78$, while *foextrap*, Dirichlet and the relaxed NSCBC are non-reflecting. A separate

	<i>foextrap</i>	Dirichlet	NSCBC-algebraic	NSCBC-LODI ($\sigma=0.25$)
R at $M = 0$	5×10^{-6}	2×10^{-6}	0.776	0.014
mean-drift residual	1.00	0.50	0.00	0.65

TABLE 1. Reflection coefficient and mean-anchoring residual (1 = no anchoring) at a quiescent boundary. No simple ghost closure achieves both low reflection and tight anchoring; the trade-off is intrinsic.

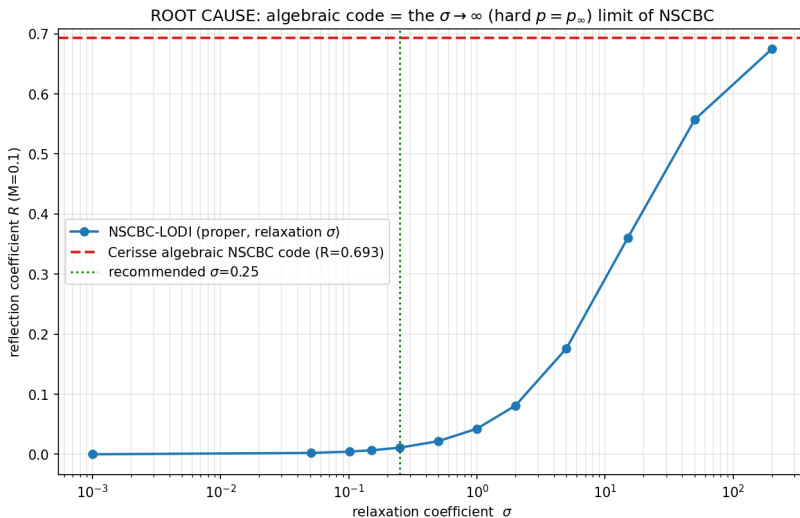


FIGURE 2. Root cause: reflection of the relaxed reference NSCBC versus its relaxation σ at $M = 0.1$. The dashed line is the as-shipped algebraic closure; it coincides with $\sigma \rightarrow \infty$.

entropy-advection test produced zero spurious pressure for all closures, confirming the differences are purely acoustic.

3.2. Root cause and correction

Figure 2 sweeps the relaxation σ of the reference (2.3) at $M = 0.1$. At $\sigma = 0$ the reference is non-reflecting; at the recommended $\sigma = 0.25$, $R = 0.011$; and as $\sigma \rightarrow \infty$ the reflection rises monotonically to $R = 0.674$ at $\sigma = 200$, converging onto the algebraic value $R = 0.693$ (dashed). *The algebraic closure is numerically the $\sigma \rightarrow \infty$ limit of a relaxed NSCBC*: it hard-pins $p = p_\infty$ (2.2) instead of relaxing toward it. This is a design choice (a fixed back-pressure outflow), not a coding error.

The correction follows directly. Rather than pinning the pressure, the subsonic-outflow branch sets the *incoming* Riemann invariant to its freestream value while keeping the outgoing invariant and entropy from the interior,

$$J_{\text{ghost}}^- = J_\infty^- = u_{n,\infty} - \frac{2c_\infty}{\gamma - 1}, \quad J_{\text{ghost}}^+ = J_{\text{int}}^+, \quad s_{\text{ghost}} = s_{\text{int}}, \quad (3.1)$$

the textbook Thompson/Whitfield characteristic non-reflecting outflow. In the 1-D test this drops R from 0.78 to below 10^{-5} at $M = 0$, while retaining the anchoring of the incoming invariant to the freestream that *foextrap* lacks. All subsequent “characteristic” (code 7) runs use (3.1).

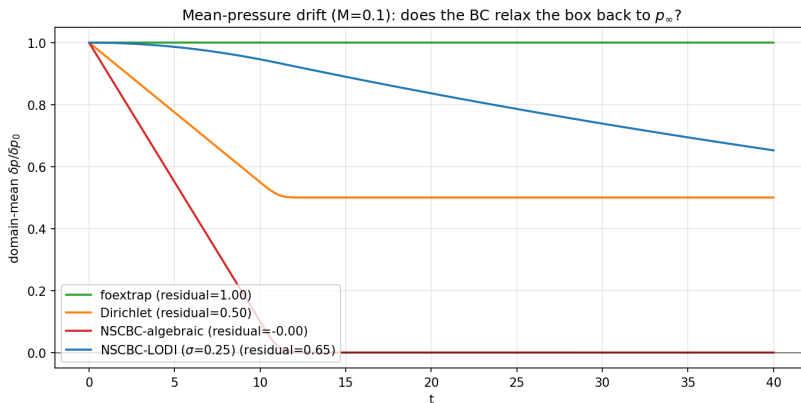


FIGURE 3. Mean-pressure relaxation from an initial over-pressure, $M = 0.1$. *foextrap* never returns to p_∞ ; the algebraic pin returns fastest. Anchoring and non-reflection are in tension.

3.3. The reflection–anchoring trade-off

Figure 3 starts the domain at a uniform over-pressure and asks whether each closure relaxes the box back to p_∞ . *foextrap* prescribes nothing, so it never anchors (residual 1.00); the algebraic pin anchors perfectly and fastest (residual 0.00) but that same pin reflects; the relaxed NSCBC sits between. The reflectivity of the algebraic closure is the price of its perfect mean-anchoring (Table 1). This tension — reflection versus anchoring — is the thread that the 2-D and shock tests resolve.

4. A two-dimensional aeroacoustic test: the co-rotating vortex pair

The 1-D benchmark isolates normal-incidence reflection. A far-field BC for aeroacoustics must also pass an obliquely radiating, multi-directional wave without corrupting it. We use the canonical model source: the sound radiated by a co-rotating vortex pair (Mitchell *et al.* 1995), solved with the production solver (2-D, WENO-Z, inviscid Euler, no IBM).

4.1. Problem and initial condition

Two identical Lamb–Oseen vortices of circulation Γ and core radius r_c sit at $\mathbf{x}_{1,2} = (\pm b/2, 0)$, each with tangential profile

$$u_\theta(r) = \frac{\Gamma}{2\pi r} \left(1 - e^{-r^2/r_c^2}\right), \quad (4.1)$$

the velocity field being their vector superposition. Being of equal sign the pair co-rotates rigidly about the origin at

$$\Omega = \frac{\Gamma}{\pi b^2}, \quad V_{\text{orb}} = \frac{1}{2}\Omega b, \quad M_{\text{orb}} = \frac{V_{\text{orb}}}{c_0}. \quad (4.2)$$

Because the configuration is invariant under rotation by π , the far field is a rotating lateral quadrupole ($m = 2$) that radiates at *twice* the orbital frequency,

$$f_{\text{ac}} = \frac{\Omega}{\pi} = 2 \cdot \frac{\Omega}{2\pi}, \quad \lambda = \frac{c_0}{f_{\text{ac}}}. \quad (4.3)$$

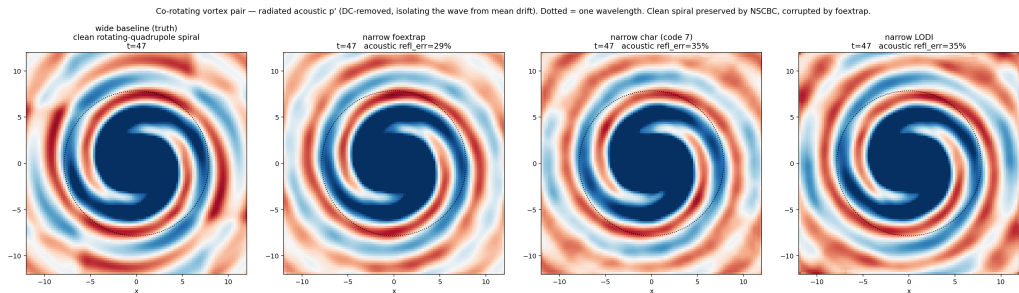


FIGURE 4. Reference (wide ± 40) radiated rotating-quadrupole spiral (pressure perturbation). This reflection-free field is the truth against which truncated-domain closures are measured.

We take $M_{\text{orb}} = 0.20$, $b = 1$, $r_c = 0.25$, $\gamma = 1.4$, with $\rho_0 = c_0 = 1$ (so $p_0 = 1/\gamma$), giving $\Gamma = 2\pi b c_0 M_{\text{orb}} = 1.257$, $\Omega = 0.40$, $T_{\text{rot}} = 2\pi/\Omega = 15.71$, $f_{\text{ac}} = 0.1273$ and $\lambda = T_{\text{ac}} = 7.854$.

The radiated pressure is only $O(\rho_0 V_{\text{orb}}^4/c_0^2) \sim 10^{-3}$, so the initial state must be in *compressible* radial equilibrium or the startup transient swamps the sound. Imposing the incompressible balance $dp/dr = \rho_0 u_\theta^2/r$ together with an isentropic density is inconsistent at $M_{\text{swirl}} \sim 0.5$ and launches a spurious monopole. The consistent state satisfies, with $p = p_0(\rho/\rho_0)^\gamma$ and $c^2 = \gamma p/\rho$,

$$\frac{dp}{dr} = \rho \frac{u_\theta^2}{r} \iff \frac{dc^2}{dr} = (\gamma - 1) \frac{u_\theta^2}{r} \implies c^2(r) = c_0^2 - (\gamma - 1) \int_r^\infty \frac{u_\theta^2(r')}{r'} dr', \quad (4.4)$$

after which $\rho = \rho_0(c^2/c_0^2)^{1/(\gamma-1)}$ and $p = p_0(c^2/c_0^2)^{\gamma/(\gamma-1)}$. The deficit integral has a closed form in exponential integrals for the Lamb–Oseen profile; it is evaluated per vortex and the two are superposed. This removes the startup monopole and leaves the clean rotating-quadrupole spiral of figure 4, computed on a wide ± 40 domain that is reflection-free over the analysis window and serves as the reference truth.

5. A drift-free, phase-aware acoustic-corruption metric

We quantify how much a truncated-domain closure corrupts the radiated wave with a single dimensionless number immune to (i) mean-pressure drift, (ii) instantaneous phase misalignment between runs, and (iii) near-field contamination.

5.1. Single-tone phasor by least squares

At each point $\mathbf{x}$ in a far-field annulus the pressure record over the window $[t_0, t_1]$ is modelled as a mean, a linear drift and the acoustic fundamental,

$$p(\mathbf{x}, t) \approx c_0(\mathbf{x}) + c_3(\mathbf{x})(t - \bar{t}) + c_1(\mathbf{x}) \cos \omega_{\text{ac}} t + c_2(\mathbf{x}) \sin \omega_{\text{ac}} t, \quad \omega_{\text{ac}} = 2\pi f_{\text{ac}}, \quad (5.1)$$

the four coefficients recovered per point by least squares on the actual (non-uniform, CFL-limited) snapshot times $\{t_k\}$. The complex quadrupole phasor is $\hat{P}(\mathbf{x}) = c_1 + ic_2$, with $p' \sim \text{Re}\{\hat{P} e^{-i\omega_{\text{ac}} t}\}$.

Why least squares and not a discrete Fourier transform. On uniform, integer-period sampling the kernel $e^{-i\omega t}$ is exactly orthogonal to the constant and linear modes, so a single-bin DFT cleanly extracts the tone. The solver's output times are non-uniform and the field carries a slow, BC-dependent mean drift; off the uniform grid the kernel is not orthogonal to $\{1, t\}$ and a naive DFT leaks the drift into the tone bin. On a synthetic record (8% time jitter plus DC, drift and tone) the integer-bin and Hann-windowed DFT

recover $|\hat{P}|$ to $\sim 1\%$ but the *complex* value (its phase) by $> 100\%$; the least-squares fit, carrying explicit $\{1, (t - \bar{t})\}$ columns, makes the tone orthogonal to the mean and drift for *any* sampling and recovers amplitude and phase to 1.5% . Neither a mean offset nor a linear drift (an un-anchored BC) can then pollute $\hat{P}$.

5.2. Drift-free broadband RMS

As a phase-insensitive companion we detrend each point (subtract the fitted $c_0 + c_3(t - \bar{t})$, not merely the mean) and form

$$\text{RMS}(\mathbf{x}) = \left(\frac{1}{T} \int_{t_0}^{t_1} [p - c_0 - c_3(t - \bar{t})]^2 dt \right)^{1/2}, \quad T = t_1 - t_0, \quad (5.2)$$

the amplitude of all unsteady content. Being an amplitude it is phase-blind by construction; detrending (not de-meaning) is essential, or a ramping mean would survive and inflate it.

5.3. Annulus, reference and corruption number

Both fields are evaluated only on $\mathcal{A} = \{\mathbf{x} : R_{\text{in}} \leq |\mathbf{x}| \leq R_{\text{out}}\}$, $R_{\text{in}} = 9.0 \approx 1.15\lambda$, $R_{\text{out}} = 10.5$. The inner edge keeps the whole annulus in the radiation zone (evanescent near-field terms, decaying faster than the radiating $\sim r^{-1/2}$ part of a 2-D multipole, are negligible); the outer edge stays 1.5 inside the narrow ± 12 boundary, clear of the truncation band. The wide ± 40 run is reflection-free in $\mathcal{A}$ over the window: a wave emitted near $t = 0$ reaches its boundary at $t \approx 40$ and re-enters $\mathcal{A}$ only at $t \approx 69.5$, after the window closes ($t_1 = 55$). For each truncated closure the corruption is the area-weighted annulus- L^2 relative error,

$$E_F^{\text{cplx}} = \frac{\|\hat{P} - \hat{P}_w\|_{\mathcal{A}}}{\|\hat{P}_w\|_{\mathcal{A}}}, \quad E_F^{\text{amp}} = \frac{\||\hat{P}| - |\hat{P}_w|\|_{\mathcal{A}}}{\||\hat{P}_w|\|_{\mathcal{A}}}, \quad E_{\text{RMS}} = \frac{\|\text{RMS} - \text{RMS}_w\|_{\mathcal{A}}}{\|\text{RMS}_w\|_{\mathcal{A}}}, \quad (5.3)$$

with $\|f\|_{\mathcal{A}} = (|\mathcal{A}|^{-1} \int_{\mathcal{A}} |f|^2 dA)^{1/2}$ and subscript w the wide truth. The window $[t_0, t_1]$ spans an integer number of acoustic periods ($4T_{\text{ac}} = [23.6, 55]$, beginning $\sim 1.5 T_{\text{rot}}$ after start) so the fit is leakage-clean.

5.4. Noise floor: the decisive honesty check

Even the reflection-free truth is imperfectly reproducible: the source is weak ($M_{\text{orb}} = 0.2 \Rightarrow$ radiated amplitude $\sim 10^{-3}$) and the annulus carries residual unsteady near field and discretisation noise. We estimate the floor by a truth-against-itself test: split the wide window into two *disjoint* halves of equal integer period count ($2 + 2T_{\text{ac}}$), compute $\hat{P}$ and RMS on each, and take their annulus- L^2 difference. **Any corruption below this floor is not resolvable.**

6. Results: the radiated wave is not a discriminator

Table 2 collects the corruption numbers for the three truncated-domain (± 12) closures — *foextrap*, corrected characteristic (3.1), and the multidimensional LODI helper with a finite, anchoring-enabling relaxation $\sigma_p = 0.25$ — against the wide truth.

6.1. The wave is below the floor

For every closure the Fourier-tone corruption (50–65%) is at or below the 64% truth-against-itself floor, and the drift-free broadband RMS corruption (5–10%) is below the

BC (narrow ± 12)	E_F^{amp} (%)	E_F^{cplx} (%)	E_{RMS} (%)	mean drift
<i>foextrap</i> (code 2)	53.7	70.3	4.8	-8.2×10^{-5}
characteristic (code 7)	49.8	85.1	5.8	$+1.4 \times 10^{-4}$
LODI ($\sigma_p=0.25$)	65.4	96.3	10.0	$+7.2 \times 10^{-3}$
noise floor (truth split)	64.1	—	14.4	—
wide (± 40 , truth)	0	0	0	4.2×10^{-4}

TABLE 2. Acoustic-corruption numbers (annulus- L^2 versus the wide truth, drift-free, phase-aware, $4T_{\text{ac}}$ window). Every closure’s wave corruption lies *at or below* the truth-against-itself noise floor (64% Fourier, 14% RMS): the radiated wave does not discriminate the closures. Only the DC mean drift separates them, where *foextrap* and characteristic anchor ($\sim 10^{-4}$) while the finite-relaxation LODI drifts $50\times$ more.

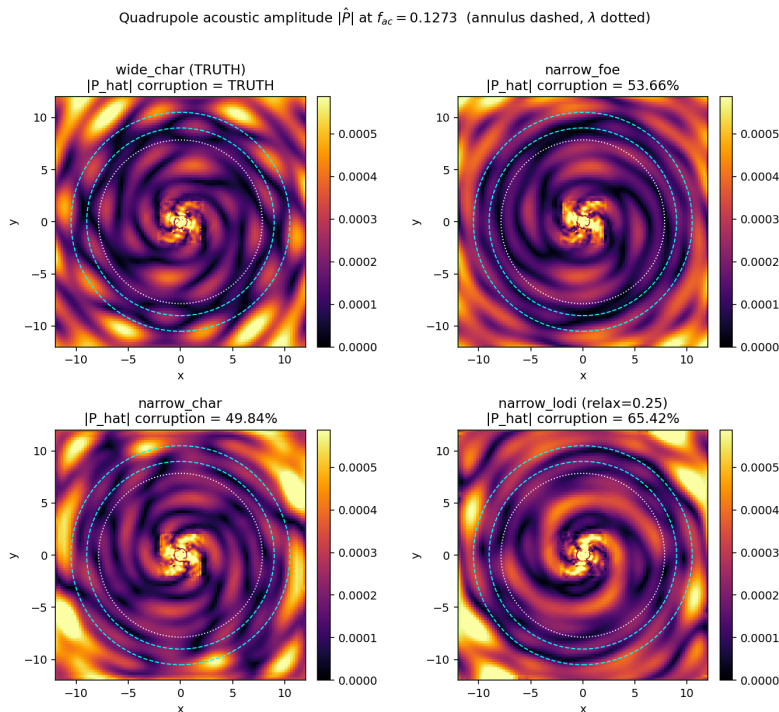


FIGURE 5. Quadrupole acoustic amplitude $|\hat{P}|$ at $f_{\text{ac}} = 0.1273$ for the four runs (annulus dashed, λ dotted). Truth and the three truncated-domain closures are visually near-identical spirals; quoted percentages are the annulus- L^2 amplitude corruptions, all near the 64% noise floor.

14% floor. The radiated quadrupole at $M_{\text{orb}} = 0.2$ is too weak, and the annulus too contaminated by the rotating-spiral near field, to resolve any closure’s effect on the wave. The large Fourier floor is the phase- L^2 sensitivity of a rotating spiral — its arms have many near-zero crossings, so a small phase shift produces a large relative L^2 . Indeed the four $|\hat{P}|$ panels (figure 5) are visually near-identical spirals, and the $r = 9$ directivity (figure 6) is a noisy, BC-independent trace rather than a clean $\cos 2\theta$ lobe pattern. The floor does *not* shrink with window length (it is set by the source’s own non-stationarity), so a longer run cannot lift the signal above it: the radiated-wave reflection is definitively not a discriminator for these closures in this case.

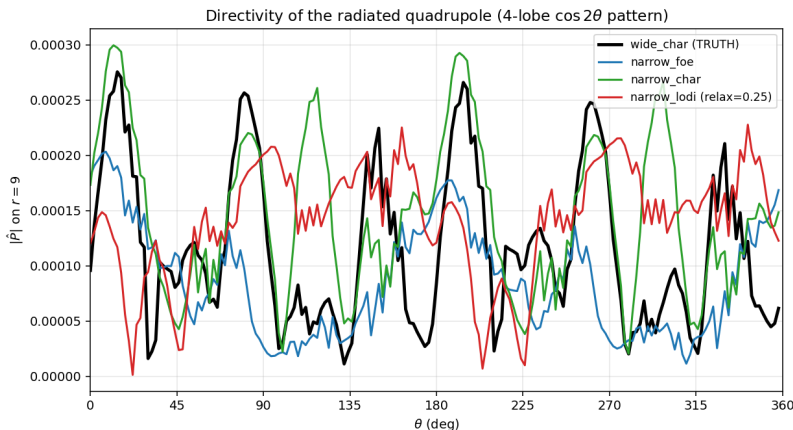


FIGURE 6. Directivity $|\hat{P}|(\theta)$ on $r = 9$. The trace is dominated by spiral-arm crossings and residual near field, not a clean $\cos 2\theta$ quadrupole lobe pattern, and the four closures trace curves of equal magnitude — visual confirmation that the wave does not discriminate them.

6.2. Mean anchoring is the only clean discriminator

The only field-level difference exceeding its noise is the DC mode — the far-field mean pressure, a weak ($< 1\%$ of p_0) effect. Over the window the annulus-mean drift is -8×10^{-5} (*foextrap*) $\approx +1.4 \times 10^{-4}$ (characteristic) $\ll +7.2 \times 10^{-3}$ (LODI), and the absolute far-field mean offset is smallest for the characteristic closure (close to the wide truth), then *foextrap*, then LODI. This reproduces the 1-D trade-off and settles the fair-LODI question: a finite relaxation $\sigma_p = 0.25$ does *not* win on the wave (it is the noisiest) and under-anchors the mean (largest residual drift). Its advertised “low reflection plus anchoring” is not realised here, because at near-normal incidence *foextrap* is already non-reflecting and the corrected characteristic anchors the mean as well, without a relaxation term. The co-rotating vortex pair is, in short, a clean *negative control*.

7. The decisive case: oblique strong-shock crossing

Where, then, does the characteristic closure earn its place? The answer is the regime the 1-D and 2-D tests cannot probe: an oblique *strong shock* crossing a near lateral boundary. We test it on a supersonic cylinder ($M_\infty = 1.7$, $\gamma = 1.4$, WENO-Z, one AMR level), holding everything fixed but the lateral BC code and progressively narrowing the lateral half-width. Cold-started, the bow shock and near field establish by $t = 30$; we measure the centreline shock stand-off Δx_s and the freestream density ρ_∞ at the upstream edge.

Figure 7 tells the story at a glance. With the characteristic closure the narrow-domain bow shock is indistinguishable from the wide baseline; with *foextrap* the upstream field is corrupted to a post-shock state (ρ_∞ too high by a factor 2.2) and the shock collapses. The reason is geometric: where an *oblique* strong shock crosses the boundary, *foextrap*’s zero-gradient extrapolation cannot represent the jump and feeds back a wrong state, whereas the characteristic closure (3.1) correctly anchors the incoming invariant to the (supersonic) freestream.

Figure 8 quantifies the gain. Taking the wide ± 8 stand-off $\Delta x_s = 1.10D$ as baseline, the characteristic closure preserves it to -2.6% at ± 2.5 , -7.1% at ± 2.0 and -18.6% at ± 1.5 , with the freestream density and post-shock ratio correct at all widths — a graceful

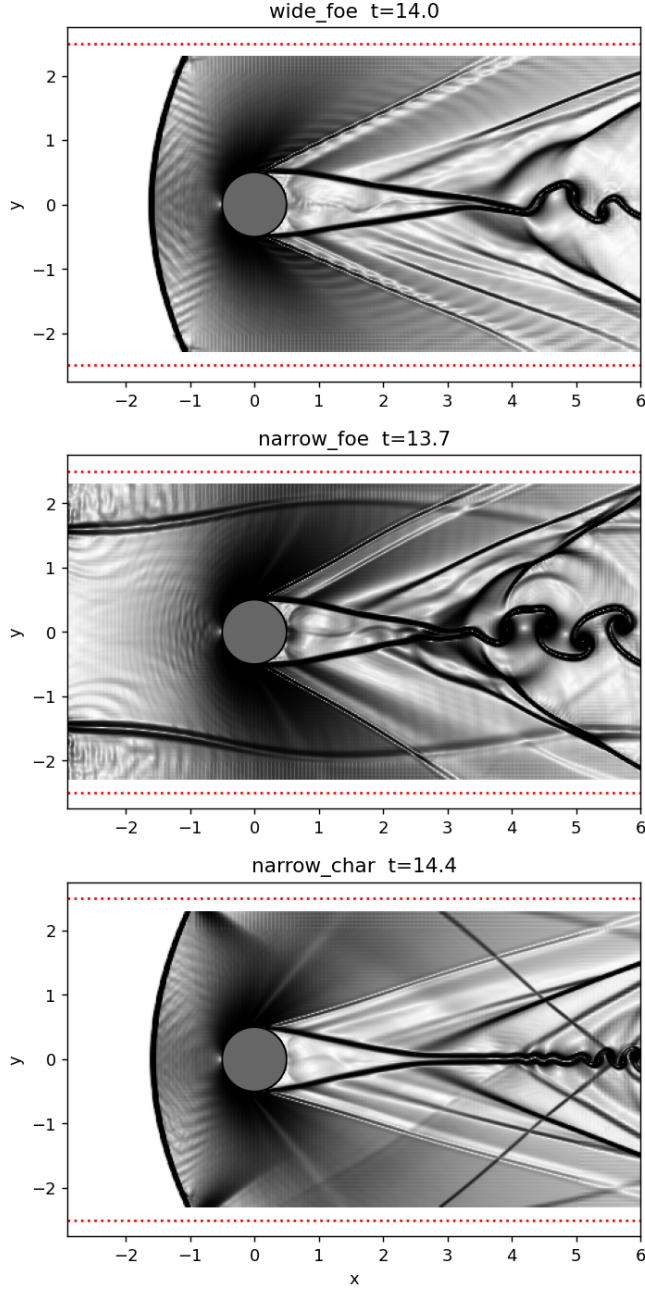


FIGURE 7. Numerical schlieren at a narrow lateral half-width (panels top to bottom). The wide baseline (top) and the corrected characteristic closure at narrow width (bottom) carry an identical bow shock; narrow *foextrap* (middle) admits a corrupted upstream state (ρ_∞ too high by $2.2\times$) and the bow shock is destroyed.

degradation, never a failure. *foextrap* is catastrophic at every narrow width. The corrected characteristic closure therefore enables a $\sim 2.5\times$ lateral-domain reduction ($\pm 5 \rightarrow \pm 2.0$) with the shock stand-off held to 7%, a regime in which *foextrap* is simply unusable. This — not free radiation — is where a characteristic far-field is decisive.

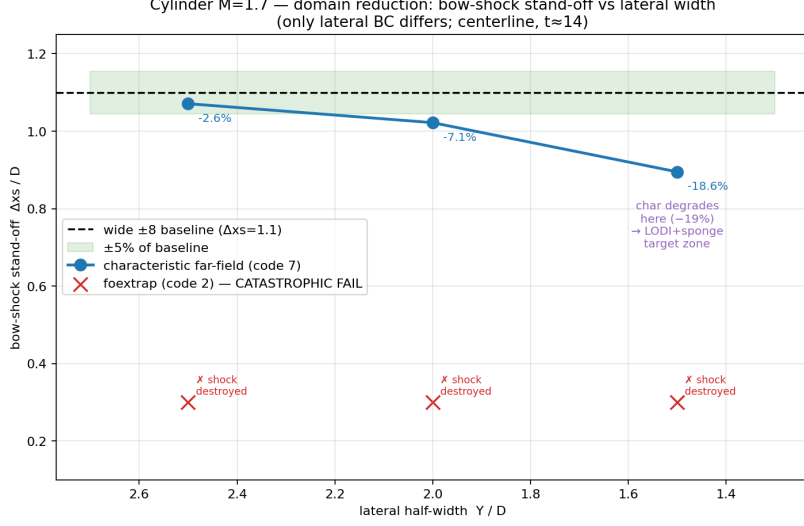


FIGURE 8. Bow-shock stand-off Δx_s versus lateral half-width (baseline wide ± 8 , $\Delta x_s = 1.10D$). The corrected characteristic closure (code 7) drifts only gradually as the boundary closes in and never fails; *foextrap* (code 2) is catastrophic at every narrow width.

8. Conclusions

Three controlled experiments, of increasing dimensionality, give a consistent picture of far-field boundary conditions for compressible immersed-boundary flow.

(i) The solver’s as-shipped algebraic “NSCBC” reflected 78% of a normally incident acoustic wave. This is exactly the $\sigma \rightarrow \infty$ limit of a relaxed characteristic condition (it hard-pins the back-pressure), not a bug. A one-line correction (3.1) — impose the incoming Riemann invariant from the freestream rather than the pressure — restores $R < 10^{-5}$ while keeping the mean-state anchoring that *foextrap* lacks.

(ii) For genuine free radiation (the co-rotating vortex pair), a rigorous drift-free, phase-aware, floor-calibrated metric shows that *no* closure beats simple extrapolation on the wave: every closure’s corruption of the radiated quadrupole lies below the truth-against-itself noise floor. The only field that discriminates the closures is the mean pressure, and even there the effect is below 1% of p_0 . A finite-relaxation LODI does not improve on this — it is the noisiest on the wave and the weakest at anchoring.

(iii) The characteristic closure is decisive in one regime the wave tests cannot probe: an oblique strong shock crossing a near boundary. There it enables a $2.5\times$ domain reduction with the bow-shock stand-off preserved to 7%, whereas *foextrap* catastrophically corrupts the upstream field.

The practical recommendation follows. On supersonic streamwise faces all closures coincide and are exact. On lateral far fields that no strong shock crosses, *foextrap* is the simplest sound choice (non-reflecting at near-normal incidence, passes entropy and vorticity, mean anchored by the supersonic inflow). Where the domain must be narrowed until a bow shock crosses the lateral boundary, the corrected characteristic closure (3.1) is required. Boundary-condition quality for compressible external flow is governed by mean-state anchoring and shock crossing — not by free-wave reflection, which is the property the non-reflecting-BC literature most often emphasises.

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